

An Accuracy Obstruction for ℓ_p Row Sparsification

A Complete-Graph Counterexample to RA-06

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Abstract

For every fixed real $p > 2$, we construct full-column-rank matrices with ordinary ℓ_p sensitivity $\mathcal{S} = \Theta_p(d)$ for which every nonnegative reweighted row subset preserving all p -th powers of norms within $1 \pm \varepsilon$ needs $\Omega_p((\mathcal{S} + d)\varepsilon^{-p})$ rows, in an explicit large-dimension regime. The matrices are grounded incidence matrices of complete graphs. An elementary, support-dependent test potential proves the lower bound. For the prescribed ordinary-sensitivity sampler, even one successful realization forces the same order of expected row count. This disproves RA-06 and the $p > 2$ quadratic-accuracy sensitivity-sampling conjecture it formalizes: no fixed polylogarithmic factor can absorb the gap between ε^{-p} and ε^{-2} .

1 Introduction and the target assertion

For a full-column-rank matrix $A \in \mathbb{R}^{n \times d}$ with rows a_i^T , its ordinary ℓ_p row sensitivities and total sensitivity are

$$s_i = \sup_{x \neq 0} \frac{|a_i^T x|^p}{\|Ax\|_p^p}, \quad \mathcal{S} = \sum_{i=1}^n s_i. \quad (1.1)$$

The question is whether these quantities alone support a row-sampling bound that is nearly linear in $\mathcal{S} + d$ and quadratic in the inverse accuracy.

Woodruff and Yasuda conjectured the sample complexity $\tilde{O}(\varepsilon^{-2}(\mathcal{S} + d))$ for sensitivity sampling [1, Section 3]. Their sampling model and ordinary-sensitivity rule appear in Definition 1.1 and Theorem 1.5 of that work. The problem catalogue entry RA-06 records the following precise $p > 2$ formulation [2].

Target assertion (RA-06). Fix $p > 2$. Do constants $C_p, c_p > 0$ exist such that, for every full-column-rank $A \in \mathbb{R}^{n \times d}$ and every $0 < \varepsilon, \delta < 1/2$, one can choose $\alpha > 0$ so that independent row retention with probabilities

$$q_i = \min\{1, 1/n + s_i/\alpha\} \quad (1.2)$$

satisfies the expected-size bound

$$\sum_{i=1}^n q_i \leq C_p \varepsilon^{-2} (\mathcal{S} + d) \log^{c_p} \left(\frac{2nd}{\varepsilon \delta} \right), \quad (1.3)$$

while the matrix $\tilde{A}$ obtained by scaling each retained row by $q_i^{-1/p}$ obeys, with probability at least $1 - \delta$,

$$(1 - \varepsilon) \|Ax\|_p^p \leq \|\tilde{A}x\|_p^p \leq (1 + \varepsilon) \|Ax\|_p^p \quad \text{for every } x \in \mathbb{R}^d? \quad (1.4)$$

We give a negative answer for every fixed real $p > 2$. The obstruction is an accuracy cost, not a claim that nearly linear sensitivity dependence is impossible at fixed accuracy. The argument has two levels. First, any nonnegative weighted subgraph approximating the complete graph's p -energy must have many support edges. Second, equal sensitivities force the rule (1.2) to use a common probability, which converts the deterministic obstruction into a lower bound on the sampler's expected size. The latter conclusion rules out every allowed choice of α , including the floor and saturation in the rule.

All logarithms are natural. Constants with a subscript p may depend on the fixed exponent p , but not on the matrix dimensions or accuracy.

2 Complete-graph matrices and exact sensitivities

Let $v \geq 3$, and write

$$[v] = \{1, \dots, v\}, \quad R = v - 1, \quad n = \binom{v}{2}, \quad d = v - 1.$$

Choose an orientation of every edge of the complete graph K_v . Let $A_v \in \mathbb{R}^{n \times d}$ be its edge-vertex incidence matrix with the column for vertex v deleted. Define the complete-graph p -energy by

$$F(z) = \sum_{1 \leq i < j \leq v} |z_i - z_j|^p, \quad z \in \mathbb{R}^v. \quad (2.1)$$

For $x \in \mathbb{R}^{v-1}$, the grounded potential $z = (x, 0)$ satisfies $\|A_v x\|_p^p = F(z)$. Conversely, replacing any $z \in \mathbb{R}^v$ by $z - z_v \mathbf{1}$ grounds its last coordinate and leaves every edge difference unchanged. Thus preservation of F for all vertex potentials is exactly the matrix embedding requirement for A_v ; grounding discards no test direction. The matrix has full column rank, since a grounded potential whose pairwise differences all vanish is zero.

Lemma 2.1 (Exact sensitivities). *For every real $p > 1$, every row of A_v has sensitivity*

$$s_v = \frac{1}{1 + (v-2)2^{1-p}}. \quad (2.2)$$

Consequently,

$$\mathcal{S} = \frac{\binom{v}{2}}{1 + (v-2)2^{1-p}} \leq 2^{p-2}v. \quad (2.3)$$

For $p \geq 2$, one also has

$$d = v - 1 \leq \mathcal{S}, \quad \mathcal{S} + d \leq (2^{p-2} + 1)v. \quad (2.4)$$

In particular, $\mathcal{S} = \Theta_p(d)$ for each fixed $p > 2$.

Proof. Fix an edge $\{i, j\}$. For each $w \notin \{i, j\}$, convexity gives

$$|z_i - z_w|^p + |z_j - z_w|^p \geq 2^{1-p} |z_i - z_j|^p.$$

Adding these inequalities, the edge $\{i, j\}$, and the remaining nonnegative energies yields

$$F(z) \geq (1 + (v-2)2^{1-p}) |z_i - z_j|^p.$$

Equality holds at $z_i = 1$, $z_j = 0$, and $z_w = 1/2$ for every other vertex. Translating this potential to grounded coordinates proves (2.2).

Summing the equal sensitivities gives the identity in (2.3). Since $1 + (v-2)2^{1-p} \geq (v-1)2^{1-p}$ for $p > 1$, its upper bound follows. For $p \geq 2$, the reverse comparison $1 + (v-2)2^{1-p} \leq v/2$ gives $\mathcal{S} \geq v - 1$. Finally, $d \leq v$ gives the last bound in (2.4). $\square$

3 A lower bound for arbitrary nonnegative row weights

Assign a finite nonnegative weight w_{ij} to each unordered edge $\{i, j\}$ of K_v , and set

$$U_w(z) = \sum_{i < j} w_{ij} |z_i - z_j|^p. \quad (3.1)$$

Write $w_{ji} = w_{ij}$ and $w_{ii} = 0$. Let H_w be the support graph, whose edges are those with positive weight. Its number of edges and the support degree of a vertex u are

$$M = |E(H_w)|, \quad \kappa_u = |\{j \neq u : w_{uj} > 0\}|.$$

Support degree counts retained edges; it is distinct from the weighted degree $d_w(u) = \sum_{j \neq u} w_{uj}$.

Theorem 3.1 (Deterministic support obstruction). *Let $p > 1$, $v \geq 3$, and $0 < \varepsilon < 1/2$. Suppose that*

$$(1 - \varepsilon)F(z) \leq U_w(z) \leq (1 + \varepsilon)F(z) \quad \text{for all } z \in \mathbb{R}^v. \quad (3.2)$$

Then every vertex has positive support degree and satisfies

$$\kappa_u^{-1/p} \leq 4\varepsilon + \frac{2\kappa_u}{R}. \quad (3.3)$$

In particular,

$$\kappa_u \geq \min\{R\varepsilon, (6\varepsilon)^{-p}\}, \quad (3.4)$$

and therefore

$$M \geq \frac{v}{2} \min\{R\varepsilon, (6\varepsilon)^{-p}\}. \quad (3.5)$$

Proof. Singleton tests control weighted degrees. Apply (3.2) to the potential equal to one at vertex u and zero elsewhere. This gives

$$(1 - \varepsilon)R \leq d_w(u) \leq (1 + \varepsilon)R \quad (u \in [v]). \quad (3.6)$$

The lower bound is positive, so $\kappa_u \geq 1$.

A support-dependent potential exposes the missing edges. Fix u , let $W = \{j : w_{uj} > 0\}$, and write $k = |W|$ and $O = [v] \setminus (W \cup \{u\})$. Choose

$$t = k^{-1/p} \in (0, 1], \quad z_u = 1, \quad z_j = t \ (j \in W), \quad z_j = 0 \ (j \in O).$$

The test is permitted to depend on H_w : the hypothesis requires preservation for every potential after the weights are fixed.

There are k edges from u to W , $R - k$ from u to O , and $k(R - k)$ from W to O . All other complete-graph edges have zero energy at this potential. Since $kt^p = 1$,

$$\begin{aligned} F(z) &= k(1 - t)^p + (R - k) + k(R - k)t^p \\ &= 2(R - k) + k(1 - t)^p. \end{aligned} \quad (3.7)$$

There are no positive-weight edges from u to O . Consequently,

$$U_w(z) = d_w(u)(1 - t)^p + t^p \sum_{i \in W} \sum_{j \in O} w_{ij}. \quad (3.8)$$

The crossing weight is at most

$$\sum_{i \in W} \sum_{j \in O} w_{ij} \leq \sum_{i \in W} d_w(i) \leq k(1 + \varepsilon)R.$$

Combining this estimate, $t^p = 1/k$, and (3.6) at u yields

$$U_w(z) \leq (1 + \varepsilon)R(1 + (1 - t)^p). \quad (3.9)$$

The energy comparison forces a large support degree. Put $h = 1 - (1 - t)^p$. The lower half of (3.2), together with (3.7) and (3.9), gives

$$2(1 - \varepsilon)(R - k) \leq (1 + \varepsilon)R(2 - h).$$

Rearranging,

$$(1 + \varepsilon)h \leq 4\varepsilon + 2(1 - \varepsilon)k/R. \quad (3.10)$$

For $p > 1$ and $0 \leq t \leq 1$, one has $(1 - t)^p \leq 1 - t$, hence $h \geq t$. It follows that

$$k^{-1/p} = t \leq 4\varepsilon + 2k/R,$$

which proves (3.3).

If k were smaller than both $R\varepsilon$ and $(6\varepsilon)^{-p}$, the right-hand side would be less than 6ε , while the left-hand side would exceed 6ε . This contradiction proves (3.4). Summing that inequality over all vertices and using $\sum_u \kappa_u = 2M$ proves (3.5). $\square$

The witness lifts every retained neighbor of u to the small common potential t . This reduces all supported edges incident to u , whereas each missing edge from u to O still has difference one in the original complete graph. Weighted-degree control limits the compensation that edges leaving W can provide. The balance $kt^p = 1$ makes the resulting error visible on the scale $k^{-1/p}$.

Corollary 3.2 (Row-subset and randomized lower bounds). *Fix $p > 2$ and $0 < \varepsilon < 1/2$. If*

$$(v - 1)\varepsilon \geq (6\varepsilon)^{-p}, \quad \text{equivalently} \quad v - 1 \geq 6^{-p}\varepsilon^{-(p+1)}, \quad (3.11)$$

then every nonnegative weighted row subset of A_v satisfying (1.4) has support size

$$M \geq \frac{6^{-p}}{2} v \varepsilon^{-p} \geq \beta_p(\mathcal{S} + d)\varepsilon^{-p}, \quad \beta_p = \frac{6^{-p}}{2(2^{p-2} + 1)}. \quad (3.12)$$

If a randomized nonnegative weighted row-subset construction succeeds with probability at least $1 - \delta$, where $0 \leq \delta < 1$, then

$$\mathbb{E}M \geq (1 - \delta) \frac{6^{-p}}{2} v \varepsilon^{-p}. \quad (3.13)$$

No independence or restriction on how the weights are chosen is required.

Proof. A weight w_{ij} corresponds to retaining the associated row of A_v scaled by $w_{ij}^{1/p}$. Grounding identifies its matrix energy with (3.1). Apply Theorem 3.1, the condition (3.11), and Lemma 2.1 to obtain (3.12). Every successful randomized output obeys the deterministic lower bound, and support size is nonnegative on the failure event; taking expectations proves (3.13). Repeated copies of a row can be combined into one nonnegative weight, so the lower bounds also apply to the total number of output rows. $\square$

4 Common-probability sampling: an expected-size obstruction

For $q \in (0, 1]$ and a subgraph $G \subseteq K_v$, define

$$U_{G,q}(z) = \frac{1}{q} \sum_{\{i,j\} \in E(G)} |z_i - z_j|^p, \quad D = qR.$$

This is the weighted energy with weight $1/q$ on each retained edge. If every edge has marginal retention probability q , the expected number of retained edges is exactly

$$\mathbb{E}|E(G)| = nq = \frac{vD}{2}. \quad (4.1)$$

Here D is a deterministic parameter, not a realized graph degree.

Theorem 4.1 (Necessary common sampling probability). *Let $p > 1$, $v \geq 3$, $0 < \varepsilon < 1/2$, and $q \in (0, 1]$. If a subgraph G satisfies*

$$(1 - \varepsilon)F(z) \leq U_{G,q}(z) \leq (1 + \varepsilon)F(z) \quad (z \in \mathbb{R}^v), \quad (4.2)$$

then

$$((1 + \varepsilon)D)^{-1/p} \leq 4\varepsilon + \frac{2(1 + \varepsilon)D}{R}, \quad (4.3)$$

and

$$D \geq \frac{1}{1 + \varepsilon} \min\{R\varepsilon, (6\varepsilon)^{-p}\}. \quad (4.4)$$

Proof. The singleton inequalities (3.6) specialize to

$$(1 - \varepsilon)R \leq \frac{\deg_G(u)}{q} \leq (1 + \varepsilon)R. \quad (4.5)$$

Thus the support degree $k = \deg_G(u)$ obeys $1 \leq k \leq (1 + \varepsilon)D$. Theorem 3.1 gives

$$((1 + \varepsilon)D)^{-1/p} \leq k^{-1/p} \leq 4\varepsilon + \frac{2k}{R} \leq 4\varepsilon + \frac{2(1 + \varepsilon)D}{R},$$

which proves (4.3). Alternatively, combining (3.4) with $k \leq (1 + \varepsilon)D$ directly gives (4.4). $\square$

Corollary 4.2 (Even one successful realization is costly). *Fix $p > 2$. Under (3.11), every common probability q that admits even one subgraph satisfying (4.2) obeys*

$$nq \geq \frac{v}{2(1 + \varepsilon)} (6\varepsilon)^{-p} \geq \frac{6^{-p}}{3} v\varepsilon^{-p}, \quad (4.6)$$

$$nq \geq \gamma_p(\mathcal{S} + d)\varepsilon^{-p}, \quad \gamma_p = \frac{6^{-p}}{3(2^{p-2} + 1)}. \quad (4.7)$$

If nq violates the first lower bound in (4.6), no subgraph can succeed; the success probability of the corresponding sampler is zero.

Proof. Use (4.4), (3.11), $nq = vD/2$, $1 + \varepsilon < 3/2$, and (2.4). The necessary inequality constrains the fixed parameter q for each individual successful graph. $\square$

Unlike (3.13), this bound has no factor $1 - \delta$. The common rescaling links every successful graph's degrees to the prescribed q , so the existence of a single successful graph already forces the expected-size bound. No realized edge count is substituted for its expectation.

5 Refutation of RA-06

By Lemma 2.1, every row of A_v has the same sensitivity. Therefore each choice of $\alpha > 0$ in (1.2) produces one common probability

$$q = \min\{1, 1/n + s_v/\alpha\} > 0. \quad (5.1)$$

Its rescaled row energy is exactly $U_{G,q}$. The results of Section 4 cover every $q \in (0, 1]$, including the saturated choice $q = 1$, and hence every probability allowed by (5.1).

Theorem 5.1 (Negative answer to RA-06). *For every fixed real $p > 2$, there are no constants $C_p, c_p > 0$ satisfying the target assertion of Section 1.*

Proof. Suppose such constants exist. For an integer $b \geq 3$, choose

$$\varepsilon = b^{-1}, \quad \delta = \frac{1}{4}, \quad v = \lceil b^{p+2} \rceil, \quad A = A_v. \quad (5.2)$$

For all sufficiently large b , one has $(v-1)\varepsilon \geq (6\varepsilon)^{-p}$, so Corollary 4.2 applies. Positive success probability requires

$$nq \geq \frac{6^{-p}}{3} vb^p. \quad (5.3)$$

On the other hand, (1.3) and Lemma 2.1 require

$$nq \leq C_p(2^{p-2} + 1) vb^2 \log^{c_p} \left(8b \binom{v}{2} (v-1) \right). \quad (5.4)$$

Since $v \leq 2b^{p+2}$ for $b \geq 3$,

$$8b \binom{v}{2} (v-1) \leq 4bv^3 \leq 32b^{3p+7}.$$

Thus there is a constant K_{p,C_p,c_p} , independent of b , such that

$$nq \leq K_{p,C_p,c_p} vb^2 (\log b)^{c_p}. \quad (5.5)$$

For every fixed $p > 2$ and fixed $c_p > 0$,

$$\frac{b^{p-2}}{(\log b)^{c_p}} \longrightarrow \infty.$$

Consequently, (5.3) and (5.5) are incompatible for large b .

Every α satisfying the proposed expected-size budget then has success probability zero, whereas the assertion requires success probability at least $3/4$. This proves the contradiction. $\square$

The matrix may depend on ε : the target quantifies over every matrix and every admissible accuracy. The sequence (5.2) also keeps n , d , and ε^{-1} polynomially related, which is why no fixed polylogarithmic factor can hide the accuracy gap.

Corollary 5.2 (The same budget fails for arbitrary reweighted subsets). *Fix $p > 2$ and a success probability $\rho \in (0, 1]$. No constants $C, c > 0$ guarantee, for every full-column-rank matrix and every $0 < \varepsilon < 1/2$, a randomized nonnegative weighted row subset satisfying (1.4) with probability at least ρ and expected support size at most*

$$C\varepsilon^{-2}(\mathcal{S} + d) \log^c \left(\frac{2nd}{\varepsilon} \right).$$

This remains false when selection is dependent and the weights are chosen adaptively.

Proof. For the sequence (5.2), (3.13) gives $\mathbb{E}M \geq \rho 6^{-p} vb^p/2$. The proposed budget is at most $K_{p,C,c} vb^2 (\log b)^c$. The same polynomial-versus-logarithmic comparison gives a contradiction. $\square$

6 Scope and relation to prior bounds

What the obstruction rules out. Theorem 5.1 refutes the joint dependence $\tilde{O}_p(\varepsilon^{-2}(\mathcal{S} + d))$ in RA-06. Corollary 5.2 goes beyond its particular sampling rule: nonnegative row reweighting and dependent selection cannot restore that budget. The lower bounds concern row-subset representations, including repeated rows after their weights are combined. They do not apply to unrestricted linear sketches whose output rows are arbitrary linear combinations of the input rows.

The dimension regime matters. The simplified ε^{-p} lower bound is asserted under (3.11); the finite- v statement is the minimum in (3.5). No ε^{-p} lower bound is claimed for all accuracies with v fixed. At fixed accuracy the construction gives a lower bound only linear in $\mathcal{S} + d$, so it does not itself rule out nearly linear sensitivity dependence in that regime. The deterministic argument holds for $p > 1$, but its comparison with quadratic inverse-accuracy dependence produces a contradiction only when $p > 2$.

Ordinary-sensitivity upper bounds. Woodruff–Yasuda’s proved ordinary-sensitivity bound is $\tilde{O}_p(\varepsilon^{-2}\mathcal{S}^{2-2/p})$ for $p > 2$ [1, Theorem 1.5]. Its larger sensitivity exponent distinguishes it from the conjectured bound rejected here. Our lower bound does not contradict that theorem. The conjecture and the sampling model are due to Woodruff–Yasuda; the deterministic support obstruction and its consequences are the results proved in this note.

A weaker inverse-accuracy dependence. Under its stated convexity and sampling-envelope hypotheses, Dai–Kosov–Murasko’s Theorem 3.5 gives a sufficient sample-size condition containing the factor $(\varepsilon^{-1} \log \varepsilon^{-1})^p$ [3]. Thus that theorem does not assert the quadratic accuracy dependence refuted here. We use it only for this comparison of accuracy exponents; no transfer of its guarantee to the particular Bernoulli rule (1.2) is asserted or needed.

References

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